# IDS Assignment 1 Solutions

## Unit 1: Data Science

## Question 1: Structured vs. Unstructured Data

**Structured data** has a predefined format and fits neatly into rows and columns within databases or spreadsheets, making it easily searchable and analyzable. This data follows a fixed schema and can be efficiently stored in relational databases. **Unstructured data**, conversely, has no predefined structure and exists in its native format without a specific organizational framework.altexsoft+2

**Three Real-World Examples of Structured Data:**

- Customer information in CRM systems including names, phone numbers, and email addressescareerfoundry
- Banking transaction data with amounts, dates, and account numberscareerfoundry
- Product databases containing prices, serial numbers, and inventory countsimperva+1

**Three Real-World Examples of Unstructured Data:**

- Social media posts, tweets, and comments on platforms like Facebook and Instagramgeeksforgeeks+1
- Email messages with attachments and multimedia contentshelf+1
- Medical images such as X-rays, MRIs, and doctor's notes in healthcare systemsshelf

**Significance in Data Analysis:**
Structured data enables straightforward statistical analysis, reporting, and automated processing due to its organized format. Unstructured data, while more challenging to analyze, contains valuable insights requiring advanced techniques like natural language processing and sentiment analysis. Organizations increasingly combine both types to gain comprehensive business intelligence and customer understanding.ibm+2

## Question 2: Four Levels of Data

The four levels of data measurement represent increasing levels of mathematical precision and analytical capability.researcher+1

**Nominal Data** categorizes information into mutually exclusive groups with no inherent order. **Example:** Types of cars (sedan, SUV, truck) or marital status (single, married, divorced). Statistical analysis is limited to frequency counts and mode calculations.scribbr+1

**Ordinal Data** orders categories but doesn't indicate magnitude differences between values. **Example:** Educational levels (high school, bachelor's, master's) or Likert scale responses (strongly disagree to strongly agree). Analysis includes median calculations and non-parametric tests.statisticshowto+1

**Interval Data** has meaningful differences between values but lacks a true zero point. **Example:** Temperature in Celsius or Fahrenheit, where zero doesn't represent absence of temperature. This enables mean calculations and parametric statistics.researcher+2

**Ratio Data** possesses all interval properties plus a true zero point representing absence of the measured attribute. **Example:** Height, weight, and age measurements where zero indicates complete absence. This allows for all statistical analyses including ratios and proportional comparisons.scribbr+1

**Impact on Statistical Analysis:**
Higher measurement levels permit more sophisticated statistical techniques, with ratio data enabling the full range of mathematical operations and parametric tests.scribbr

# Question 3: Sigma Technologies Case Study

Sigma Technologies demonstrates comprehensive data science implementation across multiple business domains. The company leverages both structured and unstructured data types to drive business decisions and operational efficiency.sigma+1

**Data Collection Methods:**
Sigma Technologies collects structured data through enterprise systems, financial transactions, and customer databases. Unstructured data sources include video content requiring transcription in 24 languages, user-generated content, and multimedia files. Their approach involves automated data sourcing combined with human annotation for complex datasets.sigmatechnology+1

**Business Value Creation:**
Data science adds value through AI-driven risk management platforms that automate detection and monitoring processes. The company uses machine learning algorithms to identify patterns in large-scale datasets, enabling predictive analytics and automated decision-making. Their data annotation services support algorithm training for major technology clients, facilitating improved search algorithms and automated query evaluation.opencorporates+2

**Operational Applications:**
Sigma implements secure data processing facilities for sensitive information handling, ensuring compliance while maintaining analytical capabilities. Their human-in-the-loop approach combines automated processing with expert validation, achieving precision requirements like 1-pixel tolerance in image annotation. This integrated methodology enables rapid scaling of data collection projects, such as gathering 1000+ dialect conversations within two months.sigma

# Unit 2: Basic Mathematics and Basic Statistics

# Question 4: Mathematical Operations

**A. Dot Product Calculation:**
For vectors **a** = (3, 4, 2) and **b** = (1, -2, 5), the dot product is calculated as :mathinsight+1

a□□b□=a1b1+a2b2+a3b3\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3a·b=a1b1+a2b2+a3b3
a□□b□=3(1)+4(−2)+2(5)=3−8+10=5\vec{a} \cdot \vec{b} = 3(1) + 4(-2) + 2(5) = 3 - 8 + 10 = 5a·b=3(1)+4(−2)+2(5)=3−8+10=5

The positive result indicates the vectors form an acute angle.mathsisfun+1

**B. Matrix Multiplication:**
Consider matrices A = [2314]\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}[2134] and B = [5123]\begin{bmatrix} 5 & 1 \\ 2 & 3 \end{bmatrix}[5213] :byjus

AB=[2(5)+3(2)2(1)+3(3)1(5)+4(2)1(1)+4(3)]=[16111313]AB = \begin{bmatrix} 2(5)+3(2) & 2(1)+3(3) \\ 1(5)+4(2) & 1(1)+4(3) \end{bmatrix} = \begin{bmatrix} 16 & 11 \\ 13 & 13 \end{bmatrix}AB=[2(5)+3(2)1(5)+4(2)2(1)+3(3)1(1)+4(3)]=[16131113]

This represents a linear transformation combining the effects of both matrices.mathinsight+1

**C. Logarithmic and Exponential Rules:**
To simplify 23x=8x+12^{3x} = 8^{x+1}23x=8x+1:
Taking logarithms: 3xlog(2)=(x+1)log(8)3x \log(2) = (x+1) \log(8)3xlog(2)=(x+1)log(8)
Since log(8)=3log(2)\log(8) = 3\log(2)log(8)=3log(2): 3xlog(2)=3(x+1)log(2)3x \log(2) = 3(x+1) \log(2)3xlog(2)=3(x+1)log(2)
Dividing by 3log(2)3\log(2)3log(2): x=x+1x = x + 1x=x+1, which gives x=−1x = -1x=−1.

# Question 5: Sampling Techniques Design

**Experimental Design:**
To study smartphone usage patterns among college students, implement both sampling methods systematically.

**Random Sampling Implementation:**

1. Obtain complete student directory (sampling frame)
2. Assign unique numbers to each student
3. Use random number generator to select 200 students
4. Contact selected participants via email/phone
5. Collect usage data through surveys or appsdatatab

**Unequal Probability Sampling Implementation:**

1. Stratify students by academic year (freshman, sophomore, junior, senior)
2. Assign higher selection probabilities to underrepresented groups
3. Weight responses during analysis to account for selection probabilities
4. Ensure minimum representation from each stratum

**Impact on Analysis:**
Random sampling provides unbiased population estimates with calculable sampling error. Unequal probability sampling may introduce bias but ensures adequate representation of subgroups, requiring weighted analysis techniques to produce valid population inferences.datatab

# Question 6: Descriptive Statistics and Empirical Rule

**Dataset:** Test scores: 78, 82, 85, 79, 88, 92, 75, 80, 86, 83

**Measures of Center:**

- **Mean:** x¯=78+82+85+79+88+92+75+80+86+8310=82.8\bar{x} = \frac{78+82+85+79+88+92+75+80+86+83}{10} = 82.8x¯=1078+82+85+79+88+92+75+80+86+83=82.8
- **Median:** Ordered data: 75, 78, 79, 80, 82, 83, 85, 86, 88, 92. Median = 82+832=82.5\frac{82+83}{2} = 82.5282+83=82.5
- **Mode:** No repeating values, so no mode exists

**Measures of Variation:**

- **Range:** 92 - 75 = 17
- **Variance:** s2=∑(xi−x¯)2n−1=29.51s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1} = 29.51s2=n−1∑(xi−x¯)2=29.51
- **Standard Deviation:** s=29.51=5.43s = \sqrt{29.51} = 5.43s=29.51=5.43

**Empirical Rule Interpretation:**
Assuming normal distribution with mean = 82.8 and standard deviation = 5.43 :geeksforgeeks+1

- **68%** of scores fall between 77.37 and 88.23 ($\mu \pm \sigma$)
- **95%** of scores fall between 71.94 and 93.66 ($\mu \pm 2\sigma$)
- **99.7%** of scores fall between 66.51 and 99.09 ($\mu \pm 3\sigma$)investopedia+1

# Unit 3: Advanced Statistics

# Question 7: Point Estimates vs. Confidence Intervals

**Point estimates** provide single-value approximations of population parameters based on sample data. **Confidence intervals** establish ranges of plausible values for population parameters with specified confidence levels.365datascience+2

**Real-World Example:**
A pharmaceutical company testing a new drug's effectiveness surveys 100 patients and finds 75% experience symptom improvement.

**Point Estimate:** 75% represents the sample proportion as a point estimate of population effectiveness.365datascience

**Confidence Interval:** With 95% confidence level, the interval might be 67% to 83%, indicating we're 95% confident the true population effectiveness lies within this range. The point estimate (75%) serves as the midpoint of the confidence interval.geeksforgeeks+1

**Practical Application:**
Point estimates provide specific values for decision-making, while confidence intervals communicate uncertainty and reliability. Wider intervals indicate greater uncertainty, while narrower intervals suggest more precise estimates.geeksforgeeks

# Question 8: One-Sample T-Test

**Dataset:** Sample weights (kg): 63, 67, 61, 69, 64, 66, 62, 68, 65, 63
**Population mean hypothesis:** μ = 65 kg

**Hypotheses:**

- **$H_0$:** μ = 65 kg (null hypothesis)
- **$H_1$:** μ ≠ 65 kg (alternative hypothesis, two-tailed test)statisticsbyjim+1

**Calculations:**

- **Sample mean:** x¯=64.8\bar{x} = 64.8x¯=64.8 kg
- **Sample standard deviation:** s = 2.57 kg
- **Sample size:** n = 10
- **Test statistic:** t=x¯−μs/n=64.8−652.57/10=−0.246t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{64.8 - 65}{2.57/\sqrt{10}} = -0.246t=s/nx¯−μ=2.57/1064.8−65=−0.246

**Assumptions:**

1. Data follows approximately normal distribution
2. Observations are independent
3. Data represents random sample from populationjmp+1

**Results Interpretation:**
With degrees of freedom = 9 and α = 0.05, the critical t-value is ±2.262. Since |t| = 0.246 < 2.262, we fail to reject $H_0$ statisticsbyjim. The sample provides insufficient evidence that the population mean differs significantly from 65 kg.

# Question 9: Chi-Square Test for Association

**Hypothetical Dataset:** Relationship between gender and product preference

| | Product A | Product B | Product C | Total |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Male | 25 | 15 | 10 | 50 |
| Female | 20 | 25 | 30 | 75 |
| Total | 45 | 40 | 40 | 125 |

**Hypotheses:**

- **$H_0$:** Gender and product preference are independentscribbr+1
- **$H_1$:** Gender and product preference are associatedstatisticssolutions+1

**Expected Frequencies Calculation:**
Expected frequency = (Row total × Column total) / Grand total

- Male-Product A: (50 × 45) / 125 = 18
- Male-Product B: (50 × 40) / 125 = 16
- Male-Product C: (50 × 40) / 125 = 16
- Female-Product A: (75 × 45) / 125 = 27
- Female-Product B: (75 × 40) / 125 = 24
- Female-Product C: (75 × 40) / 125 = 24

**Chi-Square Calculation:**
χ2=∑(Oi−Ei)2Ei=8.82\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i} = 8.82χ2=∑Ei(Oi−Ei)2=8.82

**Results Interpretation:**
With df = (2-1)(3-1) = 2 and $\alpha = 0.05$, critical value = 5.991. Since $\chi^2 = 8.82 > 5.991$, we reject $H_0$ and conclude that gender and product preference are significantly associated.

**Type I and Type II Errors:**

- **Type I Error:** Concluding association exists when variables are actually independent (false positive)
- **Type II Error:** Failing to detect association when variables are actually related (false negative)

In this context, Type I error would mean incorrectly targeting marketing based on gender when preferences are actually unrelated, while Type II error would mean missing genuine gender-based preference patterns.